

Peripheral Biorthogonalization of Critical-Branch Returns at a Quadratic Band-Merging Map:

Gauge Invariance, Branch-Memory Collapse, and a No-Go Theorem
for Normalization-Based Half Weights

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

At the first algebraic band-merging parameter of $f_u(x) = 1 - ux^2$, a two-branch Gaussian return has an almost rank-one bright channel. In the exchange-symmetric limit, both a unit-modulus cubic phase and a positive half-weight reproduce the modulus of one branch. The previous analysis left open whether the half-weight could be derived from Perron/parity-biorthogonal Grushin entrance and exit maps. We give a negative answer to that specific mechanism and identify a separate conditioning barrier.

Let R, L collect biorthonormal right and left peripheral modes, and put $Q = I - RL^*$. For raw trial and test maps V_0, W_0 , we prove that

$$G = W_0^* Q V_0, \quad V = Q V_0, \quad W^* = G^{-1} W_0^* Q$$

give the canonical peripheral pair whenever G is invertible. Then $W^* V = I$, $Q V = V$, $W^* Q = W^*$, and $V W^*$ is an idempotent packet projection inside the bulk complement. Every invertible change of biorthogonal coordinates acts on the reduced matrix by similarity. It preserves the endpoint operator, determinant, and nonzero spectrum and therefore cannot divide a bright eigenvalue by two.

The dual to the unnormalized symmetric branch vector is indeed $\frac{1}{2}(1, 1)$ under exact exchange symmetry. Together with the synthesis vector $(1, 1)^T$, however, it forms the bright projector $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, which acts as the identity on the bright range. Inserting one half into the physical return is an attenuation, not a coordinate normalization. We also show that real, static peripheral data produce a real bright reduction; a nonreal cubic multiplier requires a spectral-parameter boundary condition or another genuinely complex structure.

For two normalized branch histories with overlap c , their synthesis has singular values $\sqrt{1 \pm c}$. Consequently every exact two-label dual satisfies

$$\|W^*\| \geq (1 - c)^{-1/2}, \quad \kappa(V^* V) = \frac{1 + c}{1 - c}.$$

Seven-noise computations show progressive branch-memory collapse. At $\sigma = 10^{-4}$, the endpoint overlap is 0.99885850 and the raw Gram condition is 1751.07. Peripheral projection increases the overlap to 0.99906159 and the condition to 2130.26; the canonical analysis norm is 32.72. At the critical slice itself, the complement Gram condition is only 1.0145 and the bright coordinate dual is (0.4999905, 0.5000095). Thus branch splitting is locally stable, but closing two independent labels is not. The correct next object is a spectral-parameter bright/dark Schur complement whose self-energy can genuinely attenuate and complexify the bright channel.

Keywords: Gaussian transfer operator; biorthogonal projection; Grushin problem; branch memory; singular value; gauge invariance; Schur complement; non-normal spectrum; quadratic map.

1 Introduction

The small-noise spectrum of a Gaussian perturbation of a quadratic band-merging map has been reduced through several increasingly local operator models. The deterministic bulk determinant has an exact endpoint pole, Gaussian noise resolves a growing cloud, and endpoint singular values supply a half-logarithmic clock [12, 14]. A boundary word then gives a time-ordered cycle, while Riccati packets and a conditioned critical profile realize that cycle inside the exact noisy operator [15, 17].

Two later tests changed the proposed completion mechanism. The first showed that the second critical preimage is an order-one channel and that a sector-free time lift only replicates physical eigenvalues by Floquet rotations [13]. The second reorganized the critical slice as an exact two-branch factorization

$$\mathcal{R}(\theta) = \mathcal{E}J_\theta\mathcal{C}, \quad \mathcal{B}(\theta) = J_\theta\mathcal{C}\mathcal{E}, \quad J_\theta = \text{diag}(1, e^{i\theta}), \quad (1)$$

and proved equality of their nonzero spectra and Fredholm determinants [16]. In a rank-one exchange-symmetric model, the only nonzero return eigenvalue is

$$\eta(1 + e^{i\theta}). \quad (2)$$

Unit-modulus branch weights preserve one-branch modulus only at $\theta = \pm 2\pi/3$. A positive half-weighted sum has the same modulus, however. Radius data cannot distinguish the two.

The most economical proposed source of the half-weight was dual normalization. A Grushin problem uses separate synthesis and analysis maps, so the dual to the unnormalized bright vector $(1, 1)^T$ is naturally $\frac{1}{2}(1, 1)$. Could this be the missing factor? The answer is no. A proper dual normalization is an idempotent coordinate projection or an invertible gauge. It cannot change a nonzero eigenvalue. The number one half appears, but it is paired with a synthesis vector of norm two in branch coordinates and causes no physical attenuation.

This no-go result is accompanied by a geometric obstruction. The two critical lobes are initially disjoint, but after they traverse the common packet cycle their normalized histories become almost parallel. An exact two-label analysis map must then reconstruct a vanishing dark difference. Its norm necessarily diverges. Perron/parity projection does not restore the lost branch memory in the computed tail; it makes the final overlap slightly closer to one.

Main results and logical status

- B1. Canonical peripheral pair.** For every invertible complement Gram matrix, we construct exact trial and dual maps inside the oblique Perron/parity complement. The associated packet projection is idempotent and compatible with the global bulk projection.
- B2. Biorthogonal gauge theorem.** Every invertible change of branch coordinates transforms the reduced matrix by similarity and leaves the endpoint return unchanged. Hence ordinary dual normalization cannot supply a factor one half in the physical return.
- B3. Bright-projector no-go theorem.** The symmetric coordinate dual is exactly a half-sum, but its synthesis–analysis product is the bright projector. On a bright invariant range it is the identity. The operator that actually halves the return is not a projection.
- B4. Static-real limitation.** Real Gaussian dynamics, real peripheral projectors, and static real trial/test maps give a real reduced bright scalar. A cubic relative phase is not produced by this algebra. This does not disprove a cubic phase in a complex spectral-parameter closing map.

- B5. Branch-memory conditioning theorem.** If normalized branch histories have overlap c , their dark singular value is $\sqrt{1-c}$ and every exact two-label dual has norm at least $(1-c)^{-1/2}$. Thus branch-memory collapse forbids a uniformly bounded two-label Grushin analysis.
- B6. Seven-noise audit.** The endpoint overlap rises from 0.65084 to 0.99886 across the archived scales, while the normalized Gram condition rises from 4.73 to 1751. These are floating-point diagnostics; no asymptotic convergence theorem is claimed.
- B7. Peripheral and gauge audit.** At $\sigma = 10^{-3}$ and 10^{-4} , exact sparse Perron/parity projectors satisfy residuals below 4.0×10^{-15} . Canonical pair identities hold below 7.4×10^{-14} and a nontrivial gauge similarity holds below 3.5×10^{-17} . Peripheral extraction does not close the remaining gap to the physical bulk edge.

The corrected implication diagram is

$$\begin{array}{l}
\text{peripheral biorthogonalization} \implies \text{exact bulk-compatible packet projection,} \\
\text{dual half-sum} \not\Rightarrow \text{half-weighted physical return,} \\
\text{real static compression} \not\Rightarrow \text{cubic branch phase,} \\
\text{branch-memory collapse} \implies \text{ill-conditioned exact dark dual,} \\
\text{spectral bright/dark Schur map} \implies ? \text{ physical effective determinant.}
\end{array} \tag{3}$$

The first four statements are proved here. The final arrow is the next operator problem.

2 Gaussian operator and peripheral complement

Let u_c be the root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \tag{4}$$

and put

$$f(x) = 1 - u_c x^2, \quad u_c = 1.543689012692076 \dots \tag{5}$$

For noise $\sigma > 0$, the row-normalized folded Gaussian kernel is

$$P_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))}{Z_\sigma(x)}, \tag{6}$$

$$Z_\sigma(x) = \int_0^1 \{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))\} dy, \tag{7}$$

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}. \tag{8}$$

Let $\mathcal{K}_\sigma$ be the corresponding backward Markov operator on $\mathcal{H} = L^2(0, 1)$ and write

$$\mathcal{T}_\sigma = \mathcal{K}_\sigma^2. \tag{9}$$

At fixed noise, $\mathcal{K}_\sigma$ is Hilbert–Schmidt and $\mathcal{T}_\sigma$ is trace class.

Let r_0, r_- and ℓ_0, ℓ_- be right and left Perron/parity modes, normalized by

$$\ell_i^* r_j = \delta_{ij}, \quad i, j \in \{0, -\}. \tag{10}$$

Collect them as

$$R = (r_0, r_-), \quad L = (\ell_0, \ell_-), \quad L^* R = \mathbf{I}_2. \tag{11}$$

The oblique peripheral and bulk projections are

$$\Pi_{\text{per}} = RL^*, \quad Q = \mathbf{I} - RL^*. \tag{12}$$

They satisfy

$$Q^2 = Q, \quad QR = 0, \quad L^*Q = 0. \quad (13)$$

Since these are spectral projectors,

$$\mathcal{K}_\sigma = r_0 \ell_0^* + \lambda_- r_- \ell_-^* + K_b, \quad K_b = Q \mathcal{K}_\sigma Q. \quad (14)$$

The projection is generally nonorthogonal. Its left and right data must therefore be carried separately [6, 9].

Let $P_0, \dots, P_{k-2}$ be the packet windows along the component cycle and let P_-, P_+ be the two disjoint critical windows. The exact branch entrance and exit maps from RH-20 are denoted

$$\mathcal{C} : \mathcal{H}_0 \longrightarrow \mathcal{H}_- \oplus \mathcal{H}_+, \quad \mathcal{E} : \mathcal{H}_- \oplus \mathcal{H}_+ \longrightarrow \mathcal{H}_0. \quad (15)$$

The positive branch-complete endpoint return is $\mathcal{EC}$ and the reversed branch monodromy is $\mathcal{CE}$. Their nonzero spectra agree by the AB – BA identity [3].

3 Canonical peripheral biorthogonalization

Let $V_0 : \mathbb{C}^m \rightarrow \mathcal{H}$ be a raw synthesis map and let $W_0^* : \mathcal{H} \rightarrow \mathbb{C}^m$ be a raw analysis map. In the numerical construction, their columns are packet profiles. Define the complement Gram matrix

$$G = W_0^* Q V_0. \quad (16)$$

Theorem 3.1 (Canonical peripheral pair). *Suppose G in (16) is invertible. Define*

$$\boxed{V = Q V_0, \quad W^* = G^{-1} W_0^* Q.} \quad (17)$$

Then

$$QV = V, \quad W^*Q = W^*, \quad (18)$$

$$L^*V = 0, \quad W^*R = 0, \quad (19)$$

$$W^*V = I_m. \quad (20)$$

The finite-rank operator

$$\mathcal{P} = VW^* \quad (21)$$

is an idempotent projection satisfying

$$\mathcal{P}^2 = \mathcal{P}, \quad Q\mathcal{P} = \mathcal{P}Q = \mathcal{P}. \quad (22)$$

Proof. Idempotence of Q gives

$$W^*V = G^{-1} W_0^* Q^2 V_0 = G^{-1} G = I_m.$$

Equations (18)–(19) follow from $Q^2 = Q$, $QR = 0$, and $L^*Q = 0$. Therefore

$$\mathcal{P}^2 = V(W^*V)W^* = VW^* = \mathcal{P},$$

and the two compatibility identities follow from $QV = V$ and $W^*Q = W^*$. $\square$

This is an oblique Petrov–Galerkin construction. The invertibility and conditioning of G are the finite-dimensional inf–sup condition [1, 2]. The theorem removes the Perron/parity modes in both right and left data before packet localization is used as a reduced coordinate system.

For an operator A with $A = QAQ$, the reduced matrix is

$$M = W^*AV. \quad (23)$$

The theorem does not assert that $\mathcal{PAP}$ approximates A in operator norm. It makes the finite projection algebra exact, so that approximation and normalization are not conflated.

4 Gauge invariance and the half-weight no-go theorem

Biorthogonal coordinates have a natural gauge freedom.

Theorem 4.1 (Biorthogonal gauge invariance). *Let $S \in \text{GL}(m, \mathbb{C})$ and set*

$$V_S = VS, \quad W_S^* = S^{-1}W^*. \quad (24)$$

Then

$$W_S^*V_S = I_m, \quad V_SW_S^* = VW^*, \quad (25)$$

and the reduced matrix transforms by similarity,

$$\boxed{M_S = S^{-1}MS.} \quad (26)$$

Consequently its characteristic polynomial, determinant, trace powers, and eigenvalues are invariant.

Proof. The first two identities follow by cancellation of S and S^{-1} . The last follows from

$$W_S^*AV_S = S^{-1}(W^*AV)S.$$

Similarity gives all spectral invariants. □

The same statement is visible directly in the exact branch factorization. If

$$\mathcal{C}_S = S^{-1}\mathcal{C}, \quad \mathcal{E}_S = \mathcal{E}S, \quad (27)$$

then

$$\mathcal{E}_S\mathcal{C}_S = \mathcal{E}\mathcal{C}, \quad \mathcal{C}_S\mathcal{E}_S = S^{-1}(\mathcal{C}\mathcal{E})S. \quad (28)$$

No reciprocal scaling of entrance and exit maps can change the endpoint return or halve its bright eigenvalue.

4.1 Why the canonical half-sum is not a path half-weight

Put

$$b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (29)$$

The coordinate dual to the unnormalized bright vector in the Euclidean metric is

$$\beta^* = \frac{1}{2}(1, 1), \quad \beta^*b = 1. \quad (30)$$

Their synthesis–analysis product is

$$P_b = b\beta^* = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_b^2 = P_b. \quad (31)$$

Theorem 4.2 (Bright-projector no-go). *Assume the branch entrance is bright and the exit annihilates the dark channel,*

$$P_b\mathcal{C} = \mathcal{C}, \quad \mathcal{E}P_b = \mathcal{E}. \quad (32)$$

Then the normalized bright quotient leaves the endpoint return unchanged:

$$\boxed{\mathcal{E}P_b\mathcal{C} = \mathcal{E}\mathcal{C}.} \quad (33)$$

By contrast,

$$\mathcal{E} \left(\frac{1}{2}I_2 \right) \mathcal{C} = \frac{1}{2}\mathcal{E}\mathcal{C}. \quad (34)$$

The insertion $\frac{1}{2}I_2$ is not a projection. Therefore the physical half-weight cannot be obtained from an idempotent bright quotient or an invertible biorthogonal gauge.

Proof. Equation (33) follows immediately from either identity in (32). Equation (34) is scalar linearity, while $(\frac{1}{2}\mathbf{I}_2)^2 = \frac{1}{4}\mathbf{I}_2 \neq \frac{1}{2}\mathbf{I}_2$. $\square$

The conclusion is narrow but decisive. A half-weight could still arise from determinant combinatorics, a Schur self-energy, or a non-conservative boundary condition. It cannot be justified by saying only that the dual of $(1, 1)^T$ is $\frac{1}{2}(1, 1)$.

For a general complement metric G , the metric dual row to b is

$$\beta_G^* = \frac{b^*G}{b^*Gb}. \quad (35)$$

If branch exchange symmetry gives

$$G = \begin{pmatrix} a & c \\ c & a \end{pmatrix}, \quad (36)$$

then (35) again gives exactly $\frac{1}{2}(1, 1)$. This explains the numerical coefficients near one half without turning them into physical path weights.

5 A static real reduction does not generate a cubic phase

Proposition 5.1 (Static-real limitation). *Suppose A, Q, V_0, W_0 are real and the complement Gram matrix is invertible. Then the canonical pair in (17) and the reduced matrix $M = W^*AV$ are real. Every real gauge preserves this property. In particular, a one-dimensional bright reduction obtained from these data is a real scalar and cannot itself supply the nonreal multiplier $e^{\pm 2\pi i/3}$.*

Proof. Matrix multiplication and inversion over an invertible real matrix remain real. The one-dimensional conclusion is immediate. $\square$

A real matrix of dimension at least two can of course possess a complex conjugate eigenpair. The proposition does not forbid that. It says that a static real dual normalization does not derive a prescribed complex *relative branch coefficient*. Such a coefficient can enter naturally after complexifying the spectral parameter in a resolvent or imposing a complex closing boundary condition. The conditional cubic-phase theorem of RH-20 remains a candidate for that later mechanism; it is neither proved nor disproved here.

6 Branch-memory collapse and the dark-dual barrier

Let $u_-, u_+ \in \mathcal{H}$ be unit vectors representing two branch histories at the same packet slice. After a harmless phase choice, assume

$$\langle u_-, u_+ \rangle = c, \quad 0 \leq c < 1. \quad (37)$$

Define the synthesis

$$V_c = (u_-, u_+) : \mathbb{C}^2 \rightarrow \mathcal{H}. \quad (38)$$

Theorem 6.1 (Exact branch-memory conditioning). *The singular values of V_c are*

$$s_b = \sqrt{1+c}, \quad s_d = \sqrt{1-c}. \quad (39)$$

Consequently

$$\kappa_2(V_c) = \sqrt{\frac{1+c}{1-c}}, \quad (40)$$

$$\kappa_2(V_c^*V_c) = \frac{1+c}{1-c}. \quad (41)$$

For every exact dual W^* satisfying $W^*V_c = I_2$,

$$\boxed{\|W^*\|_2 \geq \frac{1}{\sqrt{1-c}}}. \quad (42)$$

Thus no uniformly bounded exact two-label analysis exists if $c \rightarrow 1$.

Proof. The Gram matrix is

$$V_c^*V_c = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix},$$

whose eigenvalues are $1+c$ and $1-c$. This proves (39)–(41). Let $e_d = 2^{-1/2}(1, -1)^T$. Since $W^*V_c = I_2$,

$$1 = \|e_d\| = \|W^*V_c e_d\| \leq \|W^*\| \|V_c e_d\| = \|W^*\| \sqrt{1-c},$$

which gives (42). $\square$

The bright history $(u_- + u_+)/\sqrt{2}$ stays of order one, while the dark history $(u_- - u_+)/\sqrt{2}$ has norm $\sqrt{1-c}$. Reconstructing the two labels separately therefore amplifies perturbations in the vanishing dark direction. This is a conditioning statement, not an assertion that the dark channel has no dynamical effect. A small vector can be multiplied by a large resolvent in a non-normal problem [8, 9].

6.1 Propagation from the critical split

Let ψ_-, ψ_+ be the normalized disjoint critical profiles. Starting at the final slice, define recursively

$$h_{k-1}^\pm = \psi_\pm, \quad (43)$$

$$h_j^\pm = P_j \mathcal{T}_\sigma h_{j+1}^\pm, \quad j = k-2, \dots, 0. \quad (44)$$

The normalized branch-memory overlap at slice j is

$$c_{\sigma,j} = \frac{|\langle h_j^-, h_j^+ \rangle|}{\|h_j^-\| \|h_j^+\|}. \quad (45)$$

At the critical slice, $c_{\sigma,k-1} = 0$ because the supports are disjoint. The endpoint value $c_{\sigma,0}$ measures how much branch identity survives one restricted cycle before closing.

6.2 Why peripheral projection need not separate merged histories

Proposition 6.2 (Stable projections preserve collapse). *Let Q_n be bounded projections with $\sup_n \|Q_n\| < \infty$. Suppose $\|u_n^+ - u_n^-\| \rightarrow 0$ and*

$$\min\{\|Q_n u_n^-\|, \|Q_n u_n^+\|\} \geq m > 0. \quad (46)$$

Then the normalized projected vectors also merge:

$$\left\| \frac{Q_n u_n^-}{\|Q_n u_n^-\|} - \frac{Q_n u_n^+}{\|Q_n u_n^+\|} \right\| \rightarrow 0. \quad (47)$$

Proof. For nonzero vectors x, y with norms at least m ,

$$\left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\| \leq \frac{2}{m} \|x - y\|.$$

Apply this with $x = Q_n u_n^-$ and $y = Q_n u_n^+$, then use $\|Q_n(u_n^- - u_n^+)\| \leq \|Q_n\| \|u_n^- - u_n^+\|$. $\square$

The proposition is conditional; uniform boundedness of the nonorthogonal spectral projections is itself an issue in a singular limit. It explains why removing two global modes does not automatically recreate local branch information already erased by the common packet dynamics.

7 Numerical protocol

The analytic theorems above require no numerical assumptions. The audit tests three model-specific questions: whether branch histories merge across the seven archived scales, whether peripheral projection reverses that merger, and whether canonical normalization changes the reduced bright eigenvalue.

We use the row-normalized folded midpoint matrices of RH-15–RH-20 with

$$n\sigma = 20.48 \tag{48}$$

and an eight-standard-deviation sparse Gaussian cutoff. The component period is

$$k = N_{\sigma}^{\text{H}} + 1, \tag{49}$$

where N_{σ}^{H} is the independently computed Hellinger half-energy rank from RH-16. Every packet window uses the fixed multiple $L = 6$.

The critical profiles are obtained by pulling back the endpoint Gaussian once and splitting the resulting profile at the physical partition $b = u_c^{-1/2}$. Both columns are then propagated by (44). No profile projection is inserted between packet masks, so the calculation retains the complete branch history inside each window.

At $\sigma = 10^{-3}$ and 10^{-4} , sparse left and right Perron/parity modes are resolved separately. The canonical pair (17) is constructed both at the critical slice and from the normalized endpoint histories. Four 2×2 cycle matrices are then formed:

- (i) raw dynamics with Euclidean critical profiles;
- (ii) raw dynamics with the canonical Q pair;
- (iii) K_{b}^2 inserted before every localization with Euclidean profiles;
- (iv) the same bulk dynamics with the canonical Q pair.

The full matrices are ordinary floating-point compressions. They are not operator-norm or pseudospectral enclosures.

8 Seven-noise branch-memory audit

Table 1: Endpoint branch-memory collapse. The Gram condition and dual lower bound use individually normalized histories. The final column is the smaller/larger eigenvalue-modulus ratio of the closed two-profile return.

σ	k	$c_{\sigma,0}$	$\kappa(V^*V)$	dual lower bound	return dark/bright
10^{-2}	3	0.650839	4.73	1.69	4.07×10^{-2}
4×10^{-3}	4	0.875033	15.00	2.83	2.00×10^{-2}
2×10^{-3}	5	0.970543	66.90	5.83	2.70×10^{-4}
10^{-3}	6	0.994092	337.52	13.01	4.01×10^{-5}
5×10^{-4}	6	0.984404	127.24	8.01	1.35×10^{-2}
2×10^{-4}	7	0.994965	396.22	14.09	3.99×10^{-3}
10^{-4}	8	0.998858	1751.07	29.60	1.45×10^{-5}

The clock staircase prevents monotonicity at every adjacent noise. The period stays at six between 10^{-3} and 5×10^{-4} , while the critical clearance-to-noise ratio changes. Nevertheless, the

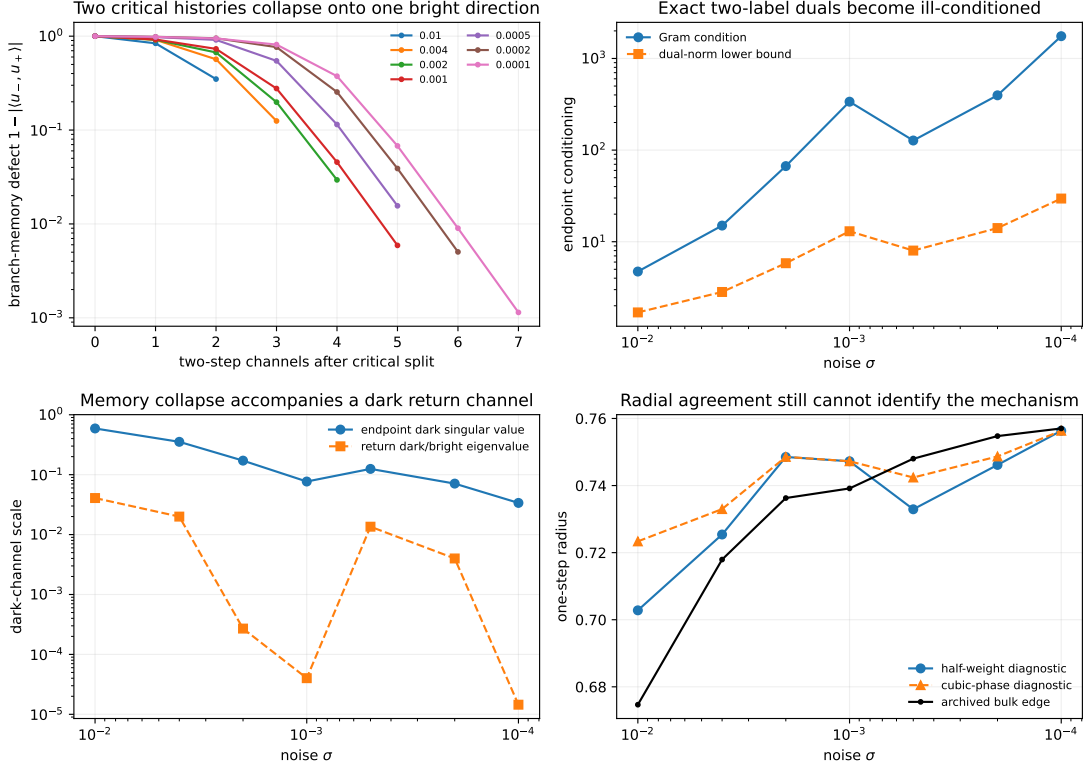


Figure 1: Branch-memory audit. Top left: the two initially disjoint critical histories merge as they traverse the common packet cycle. Top right: the endpoint Gram condition and universal dual-norm lower bound. Bottom left: the endpoint dark singular value and the dark/bright eigenvalue ratio of the closed return. Bottom right: half-weight and cubic-phase radial diagnostics remain nearly indistinguishable at branch-symmetric tail points.

overall pattern is strong. Each additional cycle depth reduces the branch-memory defect, and the cleanest smallest-noise point has a dark history singular value

$$\sqrt{1 - c_{10^{-4},0}} = 0.033786 \dots \quad (50)$$

Any exact two-label dual must therefore have norm at least 29.60 before peripheral projection.

The last column of [table 1](#) is not determined solely by $c_{\sigma,0}$. It also depends on the final closing channel and branch norm asymmetry. Its small values at the most symmetric clock cells are consistent with the same rank-one bright collapse, but no universal exponent is fitted.

The positive half-weight and cubic-phase scalar diagnostics at the smallest noise are

$$\rho_{1/2} = 0.7563353896, \quad \rho_{\text{cub}} = 0.7563358794, \quad \rho_{\text{bulk}} = 0.7570230790. \quad (51)$$

Their difference is below 5×10^{-7} . This excellent radial degeneracy is exactly why the gauge theorem is needed: a numerical radius cannot turn a coordinate half-sum into a physical attenuation.

9 Peripheral biorthogonal audit

9.1 The critical pair is stable

At the critical slice the raw profiles have disjoint support. Peripheral projection introduces a small overlap but leaves the complement Gram matrix well conditioned.

Table 2: Canonical peripheral pairs at two representative noises. The critical coefficients are the metric bright dual in (35); the endpoint quantities use normalized projected histories.

σ	$\kappa(G_{\text{crit}})$	β_-	β_+	$c_{Q,0}$	$\kappa(G_{Q,0})$	$\ W_{Q,0}^*\ $
10^{-3}	1.04594	0.499942	0.500058	0.995185	414.38	14.42
10^{-4}	1.01450	0.499991	0.500009	0.999062	2130.26	32.72

At $\sigma = 10^{-4}$, the critical complement metric is

$$G_{\text{crit}} = \begin{pmatrix} 0.99282632 & -0.00715497 \\ -0.00713512 & 0.99288349 \end{pmatrix}. \quad (52)$$

Its slight nonsymmetry comes from the oblique left/right peripheral projection. Equation (35) gives

$$\beta_G^* = (0.49999053, 0.50000947). \quad (53)$$

This is compelling evidence for an exchange-symmetric *coordinate* dual. By theorem 4.2, it is not evidence that the physical return contains the attenuation $\frac{1}{2}\text{I}_2$.

9.2 Peripheral extraction does not restore branch memory

The endpoint histories become still more parallel after Q is applied. At $\sigma = 10^{-4}$,

$$c_{\text{raw}} = 0.99885850, \quad \kappa(G_{\text{raw}}) = 1751.07, \quad (54)$$

$$c_Q = 0.99906159, \quad \kappa(G_Q) = 2130.26. \quad (55)$$

The universal lower bound for the normalized projected dual is 32.64 and the canonical analysis norm is 32.72. Thus the implementation is close to the best possible norm for this trial space; the large value is geometric, not a poor inversion algorithm.

The projected critical synthesis retains 99.76% of its norm in the two critical windows at $\sigma = 10^{-4}$. The projected endpoint histories retain 89.69% in the endpoint window. Global projection introduces visible tails, but the branch-memory conclusion is not caused by complete delocalization.

9.3 Reduced cycle radii and gauge check

Table 3: One-step radii of four static two-profile cycle compressions. “Q pair” means the canonical peripheral synthesis/analysis maps. “Bulk” means K_{b}^2 is applied before every packet localization.

σ	raw	raw + Q pair	bulk	bulk + Q pair	physical bulk
10^{-3}	0.791254	0.783898	0.761115	0.761319	0.739142
10^{-4}	0.789706	0.788348	0.782483	0.782493	0.757023

Peripheral extraction changes the selected subspace and can therefore change a compressed eigenvalue. Once the subspace is fixed, however, the gauge theorem is exact. For the nontrivial matrix

$$S = \begin{pmatrix} 1.7 & 0.25 \\ -0.35 & 0.8 \end{pmatrix}, \quad (56)$$

the largest observed error in $M_S = S^{-1}MS$ is 3.47×10^{-17} . All static reduced matrices have zero imaginary entry to floating-point precision. The largest canonical pair residual is 7.33×10^{-14} .

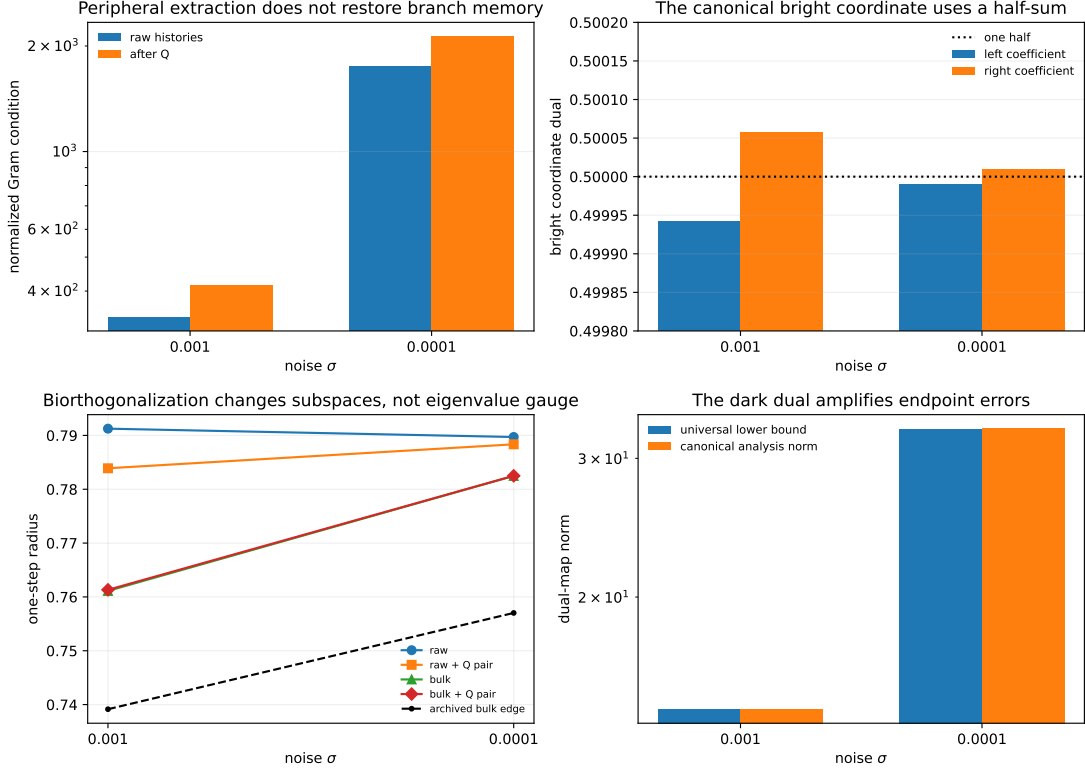


Figure 2: Peripheral-biorthogonal audit. Top left: endpoint branch-memory conditioning before and after Q . Top right: the critical bright coordinate dual is nearly the exact half-sum. Bottom left: static compressed cycle radii remain above the physical bulk edge. Bottom right: the canonical dark-dual norm nearly saturates the universal lower bound.

At the smallest noise, the bulk biorthogonal radius remains

$$0.782493 - 0.757023 = 0.025470 \quad (57)$$

above the archived physical edge. Static peripheral biorthogonalization does not solve the physical determinant problem.

9.4 Numerical status and reproducibility

The implementation tests oblique complement idempotence, canonical left/right annihilation, gauge similarity, the bright-projector no-go identity, exact merge singular values, branch-history propagation, and cycle compression. The audit records all intermediate overlaps, norms, Gram conditions, reduced matrices, residuals, software versions, source hashes, and inherited-data hashes. NumPy, SciPy, and Matplotlib provide the numerical implementation [4, 5, 10]. The largest sparse grid has $n = 204800$ midpoint cells. No interval arithmetic, directed rounding, operator-norm enclosure, or non-normal resolvent bound is used.

10 The corrected next operator target

The static normalization gate gives a definite answer. A canonical peripheral pair exists and is stable at the critical split, but it neither halves the bright return nor generates a complex phase. Carrying two exact labels around the cycle is also increasingly ill-conditioned. The next object

should therefore eliminate the dark direction dynamically rather than invert it as a permanent coordinate.

Let an exact or controlled branch effective matrix in the bright/dark basis be

$$M(z) = \begin{pmatrix} M_{bb}(z) & M_{bd}(z) \\ M_{db}(z) & M_{dd}(z) \end{pmatrix}. \quad (58)$$

Whenever $z - M_{dd}(z)$ is invertible, the scalar bright Schur map is

$$F_b(z) = z - M_{bb}(z) - M_{bd}(z)\{z - M_{dd}(z)\}^{-1}M_{db}(z). \quad (59)$$

This operation is not a coordinate gauge. Its self-energy is genuinely z -dependent, can attenuate the real bright return, and becomes complex on complex contours. It is therefore capable in principle of producing effects that [theorems 4.1](#) and [4.2](#) and [proposition 5.1](#) rule out for a static normalization [\[7\]](#).

A viable next analysis has four parts.

- D1. Balanced bright/dark data.** Use the well-conditioned critical profiles to define bright and dark coordinates. Do not reconstruct two nearly parallel endpoint histories as independent persistent labels.
- D2. Dark-channel Schur bound.** Estimate the three dark blocks in [\(59\)](#) in a norm that includes the singular factor $\sqrt{1 - c_{\sigma,0}}$. Small dark vectors alone are insufficient; the dark resolvent must be controlled.
- D3. Spectral-parameter closing law.** Determine whether the resulting scalar self-energy is asymptotically real attenuation, a complex phase on the relevant contour, or neither. No coefficient should be inserted from a radial fit.
- D4. Physical determinant comparison.** Compare the zeros of $F_b(z)$ with the parity-extracted determinant, with the complement factor and Floquet sector specified explicitly.

This target is narrower than a full complement-resolvent theorem. It first tests whether the only dynamically weak local direction—the dark branch memory—supplies the missing self-energy. Failure would rule out the entire two-branch repair rather than merely another normalization choice.

11 Discussion

11.1 What the no-go theorem changes

The half-weight versus cubic-phase ambiguity from RH-20 was partly semantic: the same coefficient one half can denote either a coordinate dual or a path attenuation. The present theorem separates them. The coordinate dual is canonical and is observed numerically to high accuracy. It forms an idempotent projector and preserves the bright eigenvalue. The path attenuation is a different operator and still needs a dynamical derivation.

Likewise, Perron/parity extraction is necessary but not sufficient. It removes the two global peripheral modes exactly, yet the static compressed cycle remains outside the physical bulk edge. This is evidence that the missing operation is a spectral self-energy or determinant cancellation, not a better normalization of the same finite matrix.

11.2 What remains encouraging

The critical two-profile space is well conditioned after peripheral projection, and its bright coordinate dual converges numerically toward the exchange-symmetric half-sum. The branch

histories then collapse onto one dominant direction, matching the very small dark return eigenvalue. This makes a scalar bright Schur problem plausible: the local finite-dimensional structure is becoming simpler, not more complicated.

The difficulty is quantitative. The dark synthesis shrinks while its exact dual grows. A successful proof must show that couplings shrink faster than the dark resolvent and dual maps grow. The seven-noise data do not establish that balance.

11.3 No arithmetic or self-adjoint implication

All operators here arise from one noisy quadratic map. The discussion of a cubic phase concerns a conditional branch-closing coefficient, not a zeta zero. No result constructs a self-adjoint Hilbert–Pólya operator, proves a prime-power trace formula, identifies the Riemann zeros, or implies the Riemann hypothesis. Any arithmetic comparison requires an independent trace identity absent from this work.

12 Conclusion

Peripheral biorthogonalization is algebraically clean but spectrally conservative. The canonical maps $V = QV_0$ and $W^* = G^{-1}W_0^*Q$ produce an exact packet projection inside the Perron/parity complement. Their gauge freedom acts by similarity and cannot halve a bright eigenvalue. Although the dual to $(1, 1)^T$ is a half-sum, its product with the bright synthesis vector is an idempotent projector that acts as the identity on the bright range. A physical half-weight is not a normalization.

The same static real construction does not derive a nonreal cubic branch coefficient. More importantly, the two branch histories lose their independence around the cycle. At the smallest computed noise their raw endpoint Gram condition is 1751, and peripheral extraction raises it to 2130. Every exact two-label dual must amplify the dark direction; the canonical map nearly saturates the sharp lower bound.

The route is again narrowed rather than closed. The stable local object is one bright channel plus a dynamically eliminated dark direction. A spectral-parameter bright/dark Schur complement is the first construction that can legitimately attenuate or complexify the bright return. Its self-energy, not a coordinate half-sum or a fitted phase, is the next gate.

Data and code availability

The source code, tests, complete CSV and JSON audits, figures, and manuscript are available with this paper [11].

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